

# ShramaSetu (श्रमसेतु)

## Track 3: Reinvent Digital Public Infrastructure (DPI) For Billions

**The Core Pitch:** India has built world-leading Digital Public Infrastructure—Aadhaar for identity, UPI for payments, and e-Shram for worker registry. Yet for 500 million informal workers, the last mile remains broken: attendance is in paper notebooks, wages are paid in cash under the table, ghost workers siphon funds, and workers have no digital credit score. **ShramaSetu is the missing 4th layer of India Stack: the Digital Public Infrastructure for Labor, Wages & Social Welfare.**

### 1. The 60-Second Stage Pitch

"Good morning judges. India has given the world the greatest Digital Public Infrastructure—Aadhaar, UPI, and e-Shram. But for 500 million unorganized blue-collar workers, the last mile of public infrastructure is completely broken. When the government issues a multi-crore highway tender, attendance is still taken in paper notebooks, wages are paid in cash under the table, ghost workers siphon funds, and workers have zero formal credit history.

For Track 3, Team Mango Bytes presents **ShramaSetu**: reimagining Digital Public Infrastructure as an open, interoperable Workforce and Wage Rail.

ShramaSetu takes passive government registries like e-Shram and turns them into active economic engines. Through AI tender extraction, digital muster rolls, transparent wage delivery directly into Jan Dhan accounts, condition-based dynamic micro-savings, and a portable ShramaID credential, we bring transparency to public works and formal financial inclusion to India's most marginalized workers."

### 2. The 4 Roles & End-to-End Workflow

| Role                                                        | What They Do in ShramaSetu                                                                                                                 | Why It Matters for Track 3 (DPI)                                                                                                      |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Employer</b><br>(PWD / NHAI / Client)                 | Uploads tender PDF; AI extracts workforce breakdown; pays tiered dynamic access fee; tracks project in <b>Project Command Center</b> .     | Enforces statutory <b>BOCW Act 1996</b> & CPWD safety norms; stops cost overruns and labor shortages in public infrastructure.        |
| <b>2. Contractor</b><br>(Workforce Operator)                | Accepts RFPs; uses 100-pt algorithm to match local workers (<15 km); logs daily digital <b>Muster Roll</b> ; disburses wages in one click. | Eliminates <b>ghost-worker fraud</b> ; replaces corrupt paper notebooks with an audit-ready, tamper-proof electronic register.        |
| <b>3. Skilled Craftsman</b><br>(Mason, Driver, Electrician) | Profile-first hiring (no English CV); uses <b>AI Career Coach</b> ; accesses Up-skilling Bridge for modern trades (Solar PV, EV charging). | Unlocks 40–60% higher wages; bridges vocational skill gaps; supports India's renewable energy and EV infrastructure targets.          |
| <b>4. Labourer</b><br>(Daily Wage Helper)                   | Receives site job invitations; tracks daily earnings transparently; saves money dynamically without starving; builds a <b>ShramaID</b> .   | Provides <b>alternative credit underwriting</b> data for banks (OCEN/Account Aggregator), bringing formal loans to unbanked citizens. |

### 3. Why Anyone Would Pay For It & Why Contractors Join (Incentives & Economics)

#### Q: "Why would Employers pay your fee?" (The Math & ROI)

- **The Ratio:** On an ₹8.2 Crore highway project, our fee is ₹15,000. That is **0.018%** (less than 2 hundredths of a percent!).
- **Project Delay Costs:** Heavy machinery (excavators, pavers, rollers) rentals cost **₹50,000 to ₹1 Lakh/day**. If a site idles for just 2 days due to missing masons or crane operators, the builder loses ₹1 to ₹2 Lakhs in rent and bank interest. Paying ₹15,000 to mobilize 100 verified workers in 48 hours saves lakhs in delay costs.
- **BOCW Compliance Protection:** Labor inspectors issue work-stoppages and fines up to **₹5 to ₹20 Lakhs** for unorganized, uninsured labor. ShramaSetu gives employers an audit-ready compliance shield.

#### Q: "Why would Contractors join instead of dealing in cash?" (The 4 Selfish Motives)

1. **Access to Lucrative Public Tenders:** Contractors cannot bid on ₹8 Crore highway tenders from tea shops. Employers on ShramaSetu send direct RFPs matched to their verified capacity.
2. **Faster Bill Release from Government:** Government PWD departments hold up contractor bills for 3 to 6 months demanding labor payment proofs. ShramaSetu auto-generates Form D / Form XII electronic muster rolls, releasing payments in **days instead of months**.
3. **Eliminating Middle-Man Mukadam Theft:** Field supervisors (Mukadams) routinely write fake ghost workers in paper registers and pocket 10% to 15% of the payroll. ShramaSetu requires worker phone confirmation, stopping supervisors from stealing contractor funds.
4. **Solving Labor Churn:** Contractors struggle with workers jumping sites for ₹50 extra. ShramaSetu contractors build a high Trust Score, attracting the most reliable crews in the district.

#### Q: "The Disintermediation Problem: Why won't they deal offline after meeting on Day 1?"

- **Construction Needs Change Constantly:** Month 1 needs earthwork helpers; Month 3 needs masons & steel fixers; Month 5 needs electricians. Contractors cannot maintain 100 permanent staff and require ShramaSetu's matching engine every new project phase.
- **Worker Refuses Cash:** Off-platform work stops the worker's ShramaID Trust Score from growing, cutting them off from formal bank loans and dynamic micro-savings.
- **Contractor Needs Audit Receipts:** If the contractor pays offline in cash, government auditors reject the labor expense under BOCW rules. Both sides have a financial incentive to stay on-platform.

## 4. Tech Stack Justification: "Why This Over That?"

| Layer         | What We Used                    | What We Rejected        | Why We Chose Our Stack (Plain English)                                                                                                                                                                                                                   |
|---------------|---------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend      | React 18 + Vite                 | Next.js (SSR) / Angular | <b>Offline PWA &amp; Zero Lag:</b> Construction sites have poor cellular data. Vite builds a pure client-side SPA that runs offline with zero server cold-start delays. Next.js SSR requires Node servers that can freeze on stage.                      |
| Styling       | Tailwind CSS                    | Material UI / Bootstrap | <b>Sunlight Legibility:</b> Workers use phones under harsh outdoor sunlight. We crafted high-contrast tokens (WCAG AAA 4.5:1) with zero runtime CSS-in-JS overhead, running at 60fps on cheap ₹6,000 phones.                                             |
| Type Safety   | TypeScript                      | Plain JavaScript        | <b>Data Integrity:</b> 4 distinct roles, tenders, muster rolls, and wage ledgers require strict data contracts, preventing undefined crashes during the presentation.                                                                                    |
| Database      | Supabase (PostgreSQL)           | MongoDB / Firebase      | <b>Relational Ledger Integrity:</b> Wage records and attendance are relational. Our schema enforces <code>unique(project_id, worker_id, date)</code> — <b>preventing double-counting of attendance or wages</b> . MongoDB lacks strict relational locks. |
| Realtime Sync | BroadcastChannel + LocalStorage | Pure Cloud WebSockets   | <b>Zero-Fail Stage Guarantee:</b> Uses browser BroadcastChannel (shramasetu_employer_sync). Employer and Contractor tabs sync across memory in 5ms <b>even if venue Wi-Fi completely dies</b> .                                                          |
| Testing       | Vitest                          | Jest                    | <b>Instant Speed:</b> Natively understands Vite; all 21 automated unit tests execute and pass in just 25 milliseconds.                                                                                                                                   |

## 5. The AI Models: Detailed Breakdown & Stage Truth

- **Google Gemini 1.5 Flash (Document Intelligence & BOQ Extraction):**
  - Government tender PDFs are 50 to 80 pages long with complex legal clauses and labor tables. Old LLMs choke on 8-page context limits.
  - Gemini 1.5 Flash has a massive **1-million-token context window**, reading the entire PDF in 2 seconds and returning structured JSON matching our `ExtractedTenderData` schema (30 Masons, 40 Helpers, 10 Electricians, 15 Operators, 5 Supervisors, BOCW safety gear, and drinking water camps).
  - Extremely low cost (\$0.075/1M tokens) and ~200ms latency, making it economically viable for Indian public works.
- **Google Cloud Translation v2 (NMT):**
  - Translates workplace phrases across 8 regional Indian languages (Hindi, Kannada, Tamil, Telugu, Malayalam, Marathi, Bengali, English).
  - Backed by an instant offline phrasebook fallback if the network is unavailable.
- **Skilled Worker AI Career Coach:**
  - Converts unstructured spoken experience into clean digital profiles, eliminating the "Resume Barrier" for blue-collar workers.
- **Honest Answer for Judges:**

"In production, we call the Gemini API with structured JSON schemas. For today's demo, to avoid conference Wi-Fi latency, we pre-loaded the extracted Belagavi Highway schema in `aiExtractionService.ts`. Our 100-point matching engine, dynamic savings math, and translation pipelines run live in real time."

## 6. The Core Algorithms & Mathematical Formulas

### A. The 100-Point Worker Matching Engine

**Score = Skill Match (50 pts) + Proximity (20 pts) + Availability (15 pts) + Experience (10 pts) + KYC (5 pts)**

- **Skill (50 pts):** Exact trade match. Handles vernacular trade aliases (e.g. *Rajmistri* or *Bricklayer* maps to *Mason*).
- **Proximity (20 pts):** Resident within project cluster (<15 km radius), preventing travel fatigue and tardiness.
- **Availability (15 pts):** Free for deployment on kickoff date (0 pts if committed to another active site).
- **Experience (10 pts):** 4+ years of verified civil/highway experience.
- **KYC / ShramalD (5 pts):** Background verified with active Aadhaar/ShramalD.

### B. The Mathematical Dynamic Savings Algorithm

**Condition Score (C) = 0.35 × Market + 0.25 × Weather + 0.25 × Safety + 0.15 × Productivity**

- **Strong Day (C ≥ 0.72):** Clear weather, high wage. Saves **3%** (e.g. ₹25). Worker capitalizes on surplus.
- **Normal Day (0.42 ≤ C < 0.72):** Moderate conditions. Saves **2%** (e.g. ₹16 on an ₹800 day).
- **Low / Risky Day (C < 0.42):** Heavy rain, low earnings. Saves **0% to 0.5%**. Protects the family's daily food budget!

*Savings sit in a liquid Jan Dhan account and can be withdrawn anytime with zero penalty.*

## 7. The 15 Brutal Judge Questions & Winning Answers

### 1. Daily wage labourers don't have smartphones and can't read English. How can they use this?

Over 75% of skilled tradesmen (masons, drivers, electricians) already carry smartphones. For unskilled helpers with basic phones or low literacy, we use an **Assisted Digital Model**: the site supervisor logs attendance on his smartphone, and the worker receives vernacular audio and SMS confirmations in their mother tongue (Hindi, Kannada, etc.). Every screen has large visual icons, and workers can tap a button to have any wage detail read aloud to them.

### 2. Contractors love cash for tax evasion and bribery. Why would they ever use ShramaSetu?

Under the BOCW Act and CPWD guidelines, government departments now legally require digital attendance records and electronic bank transfer receipts before releasing funds. Contractors who use our app get paid faster by the government, eliminate ghost-worker theft on their own sites, and retain skilled workers who refuse to work for dishonest contractors.

### 3. What prevents a corrupt contractor from adding 50 'ghost workers' (fake names) to steal money?

Two-way verification and unique ShramalDs. Every worker must have an active ShramalD verified with Aadhaar KYC. When the contractor marks attendance on site, the worker's device receives an instant confirmation. In our production build, attendance is locked to **GPS geo-fencing** (within 15 km project perimeter) and biometric Aadhaar FaceRD scanning.

### 4. Workers live hand-to-mouth. If you deduct savings, won't they starve?

That is exactly why we rejected flat savings deductions. Our **Mathematical Dynamic Savings Algorithm** checks daily conditions. On a rainy, dangerous, or low-income day, the savings rate automatically drops to **0% or ₹0**. We only recommend saving 2% to 3% on sunny days when daily earnings are high. Savings are 100% voluntary and liquid.

### 5. Do you hold workers' money in escrow? Are you an RBI-licensed Payment Aggregator?

No. ShramaSetu operates as a **Technology Service Provider (TSP)**. We never hold client funds on our balance sheet. In production, escrow and wage payouts are executed through partnerships with RBI-licensed Escrow Banking partners (like ICICI Escrow or RazorpayX). Funds move directly from the employer's nodal bank account to the worker's bank account upon milestone verification.

### 6. What about banking transaction fees and minimum balance fines on Jan Dhan accounts?

Under the Pradhan Mantri Jan Dhan Yojana (PMJDY) and RBI guidelines, basic savings bank accounts have zero minimum balance requirements and zero maintenance fees. All transfers on ShramaSetu are routed over the UPI network, which carries **zero transaction fees**.

### 7. The Government already built e-Shram with 30 crore registrations. Why does ShramaSetu need to exist?

e-Shram is a **passive database**—it sits idle in a computer. It does not find workers jobs, does not record daily site attendance, and does not disburse wages. ShramaSetu is the **active execution engine**. We don't compete with e-Shram; we integrate with it via API! We take a worker's static e-Shram identity and turn it into an active, wage-earning ShramalD.

### 8. Is ShramaSetu a closed private platform, or an open, interoperable DPI protocol?

We architected ShramaSetu as an **open protocol inspired by ONDC and Beckn**. Any contractor software, government portal, or labor union registry can connect via open APIs. A worker's ShramalD is open and portable—they can carry their verified trust score and work history to any employer or bank across India.

### 9. How does this help informal workers get bank loans without a CIBIL score?

This is where ShramaSetu connects to India's **Account Aggregator (AA)** and **Open Credit Enablement Network (OCEN)**. Traditional banks reject construction laborers because they have no formal salary slips. By using ShramaSetu, a laborer automatically builds a verified, tamper-proof record of daily earnings and attendance. Banks can use this as alternative underwriting data to offer micro-credit at fair bank interest rates instead of 50% street loan shark rates.

### 10. How do you protect worker privacy under India's DPDP Act 2023?

A worker's full phone number and personal contact details are masked until an invitation is mutually accepted. Contractors only see verified trade credentials and trust scores. All wage and attendance sharing with banks requires explicit, one-time electronic consent via a digital consent artifact.

### 11. Why not Urban Company, Apna, or WorkIndia?

Urban Company takes a 20% to 25% commission from urban gig workers (salons, AC cleaning) and doesn't handle civil infrastructure. Apna and WorkIndia are white-collar job classified boards that require written English resumes. ShramaSetu is **Work-Order Integrated, Profile-First, and Tender-Driven**.

### 12. How does Team Mango Bytes make money?

We follow a strict ethical principle: **We never charge poor laborers or take a cut from their daily wages**. Our monetization is strictly B2B: (1) Tender access fees (₹4,000 to ₹15,000 + GST); (2) Contractor SaaS subscriptions for automated BOCW filings; (3) Institutional referral commissions for micro-insurance and credit products underwritten by ShramaID.

### 13. What happens if a worker leaves mid-project or does poor work?

Their **Workforce Trust Score** on their ShramaID drops. Just as an Uber driver with a poor rating gets fewer ride requests, a worker with unexcused absences or poor feedback sees their score decrease, reducing their priority matching score for future high-wage infrastructure projects.

### 14. What is the Up-skilling Bridge and AI Coach?

In our Skilled Worker dashboard, our AI Coach helps workers transition into emerging high-demand trades—like Solar PV rooftop installation and EV charging station maintenance. This helps them increase their daily earnings by 40% to 60%.

### 15. Can you prove your matching algorithm and code actually work?

Yes! We wrote **21 automated unit tests using Vitest** in `shramasetu_core.test.ts`. All 21 tests pass in 25 milliseconds. The tests prove mathematically that an exact skill match, proximity within 15 km, and availability yield a 90+ score.

## 8. The 3-Minute Live Stage Demo Script ("The Golden Path")

### 1 Click Employer (The Client View)

- Show the **"Belagavi Highway Construction"** tender (₹8.2 Crore).
- Point out: *"Our AI analyzed this public tender and extracted 100 workers across 5 trades under BOCW statutory safety rules."*
- Point out the Dynamic Fee: ₹15,000 + 18% GST = ₹17,700.
- Click **Project Command Center** → Show the live 88% workforce fulfillment ring.

### 2 Show the 100-Point Smart Matching

- Inside Command Center, click **Smart Match Workers**.
- Show how local Masons in Belagavi score 96 and 98 points based on proximity, trade match, and availability.

### 3 Switch Persona → Contractor (Kumar Construction)

- In the Top Bar, click **Switch Persona** → Select Contractor.
- Show incoming RFP for Belagavi Highway → Click **Accept**.
- Open **Attendance Muster** → Click **"Mark All Present"**.
- Open **Wages Ledger** → Click **"Disburse All Wages"**.
- Say: *"With one click, attendance is locked, and ₹14,200 is disbursed directly to worker Jan Dhan accounts over UPI—eliminating cash skimming and ghost workers."*

### 4 Switch Persona → Labourer (Suresh Patel) / Skilled Worker (Ravi Kumar)

- Switch to Labourer.
- Show the live alert: *"Wage credited: ₹850 today!"*
- Open **Savings & Goals** → Show the **Dynamic Savings Engine**.
- Explain: *"DPI must match real human lives. On a good day, it suggests saving 3%. On a stormy monsoon day, it drops to ₹0 so the family has grocery money."*
- Open **ShramaID** → Show the digital card (SHR-RK-4821) and 94/100 Trust Score.

### 5 The Grand Finale (Multilingual Voice Translation)

- Tap the floating AI circle in the bottom-right corner.
- Switch to **Translator** → Pick English to Kannada or Hindi.
- Translate: *"What is the daily wage?"* → Show the instant translation.
- Conclude: *"ShramaSetu completes India's Digital Public Infrastructure for the 500 million workers who build our nation. Thank you!"*

## 9. Traps to Avoid on Stage

- **✗ Do NOT create a new account on stage:** Use the pre-existing profiles (Suresh Patel, Ravi Kumar, Rajesh Kumar) so all invitation notifications and wage records trigger smoothly.
- **✗ Do NOT click secondary screens:** Avoid clicking into "InsuranceScreen" or "PerformancePayScreen" unless asked. Stick strictly to the 5-step demo flow above.
- **✗ If the "Wellbeing Alert" pops up:** Simply click **"Got it"**. If a judge asks, say: *"Under the BOCW Act, site sobriety and mental focus around heavy equipment are mandatory safety requirements."*